

forma de série geométrica: $\sum_{n=0}^{\infty} a \cdot x^n = \frac{a}{1-x}$, $|a| < 1$ $x < 1$ converge $x > 1$ diverge

1a) $f(x) = \frac{1}{1+x} = \frac{1}{1-x} = \frac{1}{1-(-x)} \therefore a=1 \Rightarrow \sum_{n=0}^{\infty} 1 \cdot (-x)^n, |1-x| < 1$ ou seja converge em $\boxed{-1 < x < 1}$

b) $f(x) = \frac{1}{1-x^2} \rightarrow \frac{1}{1-x^2} = \sum_{n=0}^{\infty} 1 \cdot (x^2)^n, |x^2| < 1$ ou seja $\boxed{-1 < x < 1}$ intervalo de convergência

c) $f(x) = \frac{1}{4+x^2} = \frac{1}{4+x^2} = \frac{1/4}{1-(x^2/4)} \rightarrow a = \frac{1}{4} \Rightarrow \sum_{n=0}^{\infty} \frac{1}{4} \cdot \left(-\frac{x^2}{4}\right)^n, \left|-\frac{x^2}{4}\right| < 1$ $\frac{|x^2|}{4} < 1$ * tenho certeza da positividade. $|x^2| < 4 \therefore |x| < 2 \rightarrow \boxed{-2 < x < 2}$

d) $f(x) = \frac{1}{x-5} = \frac{1}{-5+x} = \frac{-1/5}{1-(x/5)} \therefore a = -1/5 \Rightarrow \sum_{n=0}^{\infty} \left(-\frac{1}{5}\right) \cdot \left(-\frac{x}{5}\right)^n, \left|-\frac{x}{5}\right| < 1 \rightarrow |x| < 5 \rightarrow \boxed{-5 < x < 5}$

e) $f(x) = \frac{3}{x^2+x-2} = \text{fazer por fração parcial}$

2a) $f(x) = \frac{1}{2-x}, a=5 \Rightarrow \frac{1}{2-x} = \frac{1}{2-(x+5-6)} = \frac{1}{-3-(x-5)} = \frac{-1/3}{1-(x-5)/3} = -\frac{1}{3} \sum_{n=0}^{\infty} \left(\frac{x-5}{3}\right)^n \left|\frac{x-5}{3}\right| < 1 \rightarrow |x-5| < 3 \rightarrow \boxed{2 < x < 8}$

b) $f(x) = \frac{3}{2x-1}, a=0$

$f(x) = \frac{3}{2x-1} = \frac{3}{-1+2x} = \frac{-3}{1-2x} = -3 \sum_{n=0}^{\infty} (2x)^n \Rightarrow f(x) = \sum_{n=0}^{\infty} -3 \cdot (2x)^n$ $|2x| < 1$ $|x| < \frac{1}{2}$ $\boxed{-\frac{1}{2} < x < \frac{1}{2}}$ Intervalo de convergência

c) $f(x) = \frac{1}{2x-5}, a=-3$

$f(x) = \frac{1}{2x-5} = \frac{1}{-5+2x} = \frac{1}{-5+2(x+3)-6} = \frac{1}{-11-(x-3)} = \frac{-1/11}{1-(x-3)/11} = \frac{-1/11}{1-(x-3)/11} = \sum_{n=0}^{\infty} \left(-\frac{1}{11}\right) \cdot \left(\frac{x-3}{11}\right)^n = \sum_{n=0}^{\infty} \left(-\frac{1}{11}\right) \cdot \left(\frac{x-3}{11}\right)^n \cdot |x-3| < 11$ $\frac{2|x+3|}{11} < 1$ $|x+3| < 11/2$ $-\frac{11}{2} < x+3 < \frac{11}{2}$ $\boxed{-\frac{17}{2} < x < \frac{5}{2}}$

d) $f(x) = \frac{3}{x+2}, a=0$

$f(x) = \frac{3}{x+2} = \frac{3}{2-(-x)} = \frac{3/2}{1-(x/2)} = \sum_{n=0}^{\infty} \frac{3}{2} \cdot \left(-\frac{x}{2}\right)^n$ $\frac{|x|}{2} < 1$ $\boxed{-2 < x < 2}$